<u>Many-particle systems</u>: bosons = symmetric wave function

fermions = antisymmetric

$$\psi(\bar{r}_1, \bar{r}_2) = \pm\, \psi(\bar{r}_2, \bar{r}_1)$$

$\Rightarrow$ Pauli principle: two identical fermions cannot occupy the same state.

<u>Hydrogenic states</u> (n, l, m) = orbitals

$$H = \sum_{j=1}^{Z} \underbrace{\left\{ -\frac{\hbar^2}{2m} \nabla_j^2 - \frac{1}{4\pi\varepsilon_0} \frac{Ze^2}{r_j} \right\}}_{\text{K.E. + P.E. of j-th electron}} + \frac{1}{2} \underbrace{\frac{1}{4\pi\varepsilon_0} \sum_{j \neq k}^{Z} \frac{e^2}{|\vec{r}_j - \vec{r}_k|}}_{\text{repulsion of j-th and k-th electrons}}$$

$$\Psi = \psi(r_1, r_2, \ldots, r_Z)\, \chi(s_1, s_2 \ldots s_Z)$$

must be antisymmetric w.r.t. interchange of any two electrons

* Neglecting $V(r_j - r_k)$ [e-e interaction], there are $2n^2$ different states (with the same E_n) for a given of $\underline{\underline{n}}$
↳ ($\uparrow, \downarrow$ spin degeneracy)

K	$n=1$ shell	2 ~~electrons~~	0	$(1s)^2$
L	$n=2$	8	0	$(2s)^2$
			1	$(2p)^6$
M	$n=3$	18	0	$(3s)^2$
			1	$(3p)^6$
			2	$(3d)^{10}$

<u>Hund's rules</u> (for the <u>ground states</u>!)

① state with highest total spin (S) has the lowest energy
symmetric spin w.f. $\Rightarrow$ ~~asym~~ antisymmetric space w.f.
$\Rightarrow$ minimizes <u>Coulomb</u>

② For a given spin, the state with highest L (consistent with antisymmetrization), will have the lowest energy.

③ If the subshell (n, ℓ) is less (or equal) than $\frac{1}{2}$-filled,

$\Rightarrow$ lowest energy when $J = |L - S|$.

If it is more than $\frac{1}{2}$-filled, than $J = L + S$

<u>Hund 1</u> minimizes Coulomb ; as does <u>Hund2</u> which suggests that Coulomb is minimized by rotating in the same direction (maximazes distance)

<u>Hund3</u> minimizes <u>spin-orbit</u> interaction

$$H_{so} = \lambda \vec{S} \cdot \vec{L}$$

with H_{so}, $\vec{L}$ and $\vec{S}$ not conserved but $\vec{J} = \vec{L} + \vec{S}$ is!

$$J \in (|L - S|, L + S)$$

$$\vec{J}^2 = (\vec{L} + \vec{S})^2 = \vec{L}^2 + \vec{S}^2 + 2\vec{L}\vec{S}$$

$$\Rightarrow \langle \lambda \vec{L} \cdot \vec{S} \rangle = \frac{\lambda}{2} \hbar^2 \left[J(J+1) - L(L+1) - S(S+1) \right]$$

<u>Hydrogen</u>

where $\lambda = \dfrac{e^2}{8\pi\varepsilon_0} \dfrac{1}{m^2 c^2 r^3}$

<u>PG 6.35</u>

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{1}{\ell(\ell + \frac{1}{2})(\ell+1)\, n^3 a^3}$$

← Note that depends only on shell [n, ℓ numbers] but not $m_\ell = l_z$!

Griffiths problem

$$\Rightarrow \frac{\lambda \hbar^2}{2} = \frac{e^2}{8\pi\varepsilon_0} \frac{1}{m^2 c^2} \frac{\hbar^2}{2} \frac{j(j+1) - \ell(\ell+1) - s(s+1)}{\ell(\ell+\frac{1}{2})(\ell+1)\, n^3 a^3}$$

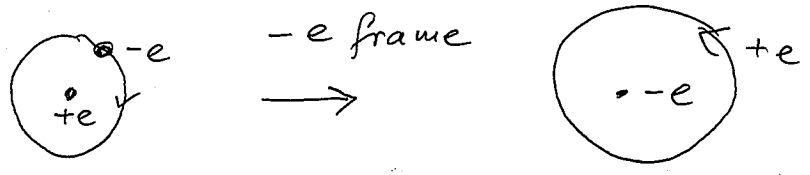
$$a = \frac{4\pi\varepsilon_0 \hbar^2}{m e^2}, \quad E_n = -\frac{\hbar^2}{2 m a^2 n^2}$$

$$\frac{1}{mc^2} \frac{\hbar^2}{2 m a^2 n^2} \underbrace{\left(\frac{e^2}{8\pi\varepsilon_0 a n} \right)}_{\frac{m e^4}{2n (4\pi\varepsilon_0)^2 \hbar^2} = -n E_n} = \frac{1}{mc^2} E_n^2 \cdot n$$

$$\Rightarrow \langle \lambda \vec{L} \cdot \vec{S} \rangle = \frac{E_n^2 n}{mc^2} \frac{j(j+1) - \ell(\ell+1) - s(s+1)}{\ell(\ell+\frac{1}{2})(\ell+1)}$$

Obviously relativistic correction!

Spin-orbit coupling



proton (+e) produces current $I = e/\tau$

$\Rightarrow$ magnetic field $\vec{B} = \frac{\mu_0 \vec{I}}{2r}$

Electron's orbital momentum $L = rmv = rm\frac{2\pi r}{\tau} = 2\pi\frac{mr^2}{\tau}$

$$\frac{1}{\tau} = \frac{L}{2\pi m r^2}$$

$$\Rightarrow B = \frac{\mu_0}{2r}\frac{e}{\tau} = \frac{\mu_0 e}{4\pi m r^3} L$$

$$\sqrt{\mu_0 \varepsilon_0} = \frac{1}{c} \Rightarrow \mu_0 = \frac{1}{c^2 \varepsilon_0}$$

$$\vec{B} = \frac{1}{4\pi\varepsilon_0}\frac{e}{mc^2 r^3}\vec{L}$$

$\Rightarrow$ electron's ~~spin~~ magnetic moment interacts with that $\vec{B}$:

$$H_{so} = -\vec{\mu}_e \cdot \vec{B} \quad , \vec{\mu}_e = \frac{g(-e)}{2m}\vec{S} = -\frac{e}{m}\vec{S} \quad (g=2)$$

$$\Rightarrow H_{so} = \frac{e^2}{4\pi\varepsilon_0}\frac{1}{m^2c^2r^3}\vec{S}\cdot\vec{L}$$ spin-orbit coupling

Caveat: the frame we used is not <u>inertial</u> $\Rightarrow$ Thomas correction gives factor $(1/2)$ — follows from Dirac's eq.

$$H_{so} = \frac{e^2}{8\pi\varepsilon_0}\frac{1}{m^2c^2r^3}\vec{S}\cdot\vec{L} = \lambda\,\vec{S}\cdot\vec{L}$$

For hydrogen-like atom of charge Ze : $\lambda = \frac{Ze^2}{8\pi\varepsilon_0 m^2c^2r^3}$

Moreover $\left\langle \frac{1}{r^3}\right\rangle = \frac{Z^3}{a_0^3 n^3 \, l(l+\frac{1}{2})(l+1)}$

$\Rightarrow \lambda \sim Z^4$: SO becomes strong in heavy atoms.

Another point of view (very closely related):

orbiting nucleus $(+e)$ creates magnetic field

$$\vec{B} = \frac{\vec{E} \times \vec{v}}{c^2}, \quad \vec{E} = -\vec{\nabla} V(r) = -\hat{r}\,\frac{dV(r)}{dr}$$ is the electric field of $(+e)$

[Griffiths E&M, Ch. 12]

$$= -\frac{\vec{r}}{r}\,\frac{dV(r)}{dr}$$

$$\vec{B} = -\frac{1}{r}\frac{dV}{dr}\,\frac{\vec{r}\times\vec{v}}{c^2} = \frac{1}{r}\frac{dV}{dr}\,\frac{m\vec{v}\times\vec{r}}{mc^2} = \frac{1}{r}\frac{dV}{dr}\,\frac{\vec{L}}{mc^2}$$

But $V(r) = -\dfrac{Ze^2}{4\pi\varepsilon_0}\dfrac{1}{r}$

$$\vec{B} = \frac{Ze^2}{4\pi\varepsilon_0 r^3}\,\frac{\vec{L}}{mc^2}$$

$$\Rightarrow \quad H_{so} = -\underbrace{\left(\tfrac{1}{2}\right)}_{\text{Thomas factor}} \vec{\mu}_e \vec{B} = \frac{Ze^2}{8\pi\varepsilon_0 m^2 c^2 r^3}\,\vec{S}\cdot\vec{L}$$ as before.

Consider atom with one not fully filled shell

$$H_{so} = \sum_a \alpha_a \vec{l}_a \vec{s}_a \qquad \text{sum is over electrons in that shell.}$$

Maximum possible number of electrons n in a given shell is $2(2l+1)$: can have 2 electrons (with spins $\uparrow$ and $\downarrow$) in every state with m_l (NOTE: $\hat{L}_z|\rangle = m_L|\rangle$).

① Suppose now that $n < \frac{1}{2} n_{max} = 2l+1$.

Then by Hund's 1ST rule $\underbrace{S_{tot} = \frac{n}{2}}_{(n=2S)}$ $\left(M_S = n \cdot \frac{1}{2}\right)$

$$\Rightarrow \vec{s}_a = \frac{1}{n} \vec{S}$$

$\vec{s}_a$: spin of a-th electron; $\vec{S}$: total spin

$$\Rightarrow H_{so} = \sum_a \frac{\alpha_a}{n} \vec{l}_a \cdot \vec{S} = \frac{\alpha}{2S} \left(\sum_a \vec{l}_a\right) \cdot \vec{S} = \frac{\alpha}{2S} \vec{L} \cdot \vec{S}$$

Here $\alpha_a = \alpha$ for electrons from <u>the same shell</u>

$$\Rightarrow \alpha = \frac{Ze^2}{8\pi\varepsilon_0 m^2 c^2 r^3} > 0.$$

Thus $H_{so}(n < 2l+1) = A\, \vec{S} \cdot \vec{L}$ with $A > 0$

lowest energy configuration will then have $\vec{S}$ antiparallel with $\vec{L}$ so that $\vec{S} \cdot \vec{L} < 0$

$$\Rightarrow J = |L - S|.$$

② Let now $n > 2l+1 = \frac{1}{2} n_{max}$.

$$H_{so} = \sum_{a=1}^{n} \alpha\, \vec{l}_a \cdot \vec{s}_a = \underbrace{\sum_{a=1}^{n_{max}} \alpha\, \vec{l}_a \cdot \vec{s}_a}_{\text{fully filled shell}} - \underbrace{\sum_{a=n+1}^{n_{max}} \alpha\, \vec{l}_a \cdot \vec{s}_a}_{\text{hole contribution}}$$

For the fully filled shell $H_{so}(\text{filled}) = 0$ because $\vec{L}\cdot\vec{S} = 0$ because $\vec{S} = 0$ than.

Hence $H_{so}(n > 2l+1) = -\alpha \sum_{a=n+1}^{n_{max}} \vec{l}_a \cdot \vec{s}_a$

To minimize Coulomb repulsion between holes we maximize total spin again $S = \frac{n}{2}$

$$\vec{S}_a = \frac{1}{n}\vec{S} = \frac{1}{2S}\vec{S}.$$

$$\Rightarrow H_{so} = -\frac{\alpha}{2S}\vec{S}\cdot\left(\sum_a \vec{l}_a\right) = -\frac{\alpha}{2S}\vec{L}\cdot\vec{S}.$$

Thus now $A = -\frac{\alpha}{2S} < 0$ and lowest energy state has $\vec{S}$ and $\vec{L}$ parallel : $\vec{L}\cdot\vec{S} > 0$

$\Rightarrow$ $\underline{J = L + S}$ here.

This explains Hund's 3rd rule:

$J = |L - S|$ for $n < \frac{n_{max}}{2}$

$J = L + S$ for $n > \frac{n_{max}}{2}$

Clearly for $n = \frac{1}{2} n_{max}$ both give the same: $J = S$

This is because in this situation every M_L has one electron and total $\hat{L}_z = 0$.

Ashcroft, Mermin: Solid State Physics

When a solid contains some ions with partially filled electronic shells, its magnetic behavior is very different. Before we can apply the general result (31.20) to this case, we must review the basic facts about the low-lying states of such ions.

GROUND STATE OF IONS WITH A PARTIALLY FILLED SHELL: HUND'S RULES

Suppose we have a free[16] atom or ion in which all electronic shells are filled or empty except for one, whose one-electron levels are characterized by orbital angular momentum l. Since for given l there are $2l + 1$ values l_z can have (l, $l - 1$, $l - 2, \ldots, -l$) and two possible spin directions for each l_z, such a shell will contain $2(2l + 1)$ one-electron levels. Let n be the number of electrons in the shell, with $0 < n < 2(2l + 1)$. If the electrons did not interact with one another, the ionic ground state would be degenerate, reflecting the large number of ways of putting n electrons into more than n levels. However, this degeneracy is considerably (though in general not completely) lifted by electron-electron Coulomb interactions as well as by the electron spin-orbit interaction. Except for the very heaviest ions (where spin-orbit coupling is very strong) the lowest-lying levels after the degeneracy is lifted can be described by a simple set of rules, justified both by complex calculations and by the analysis of atomic spectra. Here we shall simply state the rules, since we are more interested in their implications for the magnetic properties of solids than in their underlying justification.[17]

1. *Russel-Saunders Coupling* To a good approximation[18] the Hamiltonian of the atom or ion can be taken to commute with the total electronic spin and orbital angular momenta, $\mathbf{S}$ and $\mathbf{L}$, as well as with the total electronic angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Therefore the states of the ion can be described by quantum numbers L, L_z, S, S_z, J, and J_z, indicating that they are eigenstates of the operators $\mathbf{L}^2$, $\mathbf{L}_z$, $\mathbf{S}^2$, $\mathbf{S}_z$, $\mathbf{J}^2$, and $\mathbf{J}_z$ with eigenvalues $L(L + 1)$, L_z, $S(S + 1)$, S_z, $J(J + 1)$, and J_z, respectively. Since filled shells have zero orbital, spin, and total angular momentum, these quantum numbers describe the electronic configuration of the partially filled shell, as well as the ion as a whole.

2. *Hund's First Rule* Out of the many states one can form by placing n electrons into the $2(2l + 1)$ levels of the partially filled shell, those that lie lowest in energy have the largest total spin S that is consistent with the exclusion principle. To see what that value is, one notes that the largest value S can have is equal to the largest magnitude that S_z can have. If $n \leqslant 2l + 1$, all electrons can have parallel spins without multiple occupation of any one-electron level in the shell, by assigning them levels with different values of l_z. Hence $S = \frac{1}{2}n$, when $n \leqslant 2l + 1$. When $n = 2l + 1$,

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16 We shall discuss how the behavior of the free atom or ion is modified by the crystalline environment on pages 656–659.

17 The rules are discussed in most quantum mechanics texts. See, for example, L. D. Landau and E. M. Lifshitz, *Quantum Mechanics*, Addison Wesley, Reading, Mass., 1965.

18 The total angular momentum J is always a good quantum number for an atom or ion, but L and S are good quantum numbers only to the extent that spin-orbit coupling is unimportant.

S has its maximum value, $l + \frac{1}{2}$. Since electrons after the $(2l + 1)$th are required by the exclusion principle to have their spins opposite to the spins of the first $2l + 1$, S is reduced from its maximum value by half a unit for each electron after the $(2l + 1)$th.

3. ***Hund's Second Rule*** The total orbital angular momentum L of the lowest-lying states has the largest value that is consistent with Hund's first rule, and with the exclusion principle. To determine that value, one notes that it is equal to the largest magnitude that L_z can have. Thus the first electron in the shell will go into a level with $|l_z|$ equal to its maximum value l. The second, according to rule 2, must have the same spin as the first, and is therefore forbidden by the exclusion principle from having the same value of l_z. The best it can do is to have $|l_z| = l - 1$, leading to a total L of $l + (l - 1) = 2l - 1$. Continuing in this way, if the shell is less than half filled, we will have $L = l + (l - 1) + \cdots + [l - (n - 1)]$. When the shell is precisely half filled, all values of l_z must be assumed, and therefore $L = 0$. The second half of the shell is filled with electrons with spin opposite to those in the first half, and therefore the exclusion principle allows us again to go through the same series of values for L we traversed in filling the first half.

4. ***Hund's Third Rule*** The first two rules determine the values of L and S assumed by the states of lowest energy. This still leaves $(2L + 1)(2S + 1)$ possible states. These can be further classified according to their total angular momentum J, which, according to the basic rules of angular momentum composition, can take on all integral values between $|L - S|$ and $L + S$. The degeneracy of the set of $(2L + 1)(2S + 1)$ states is lifted by the spin-orbit coupling, which, within this set of states, can be represented by a term in the Hamiltonian of the simple form $\lambda(\mathbf{L} \cdot \mathbf{S})$. Spin-orbit coupling will favor maximum J (parallel orbital and spin angular momenta) if λ is negative, and minimum J (antiparallel orbital and spin angular momenta) if λ is positive. As it turns out, λ is positive for shells that are less than half filled and negative for shells that are more than half filled. As a result, the value J assumes in the states of lowest energy is:

$$\begin{aligned} J &= |L - S|, \qquad n \leqslant (2l + 1), \\ J &= L + S, \qquad n \geqslant (2l + 1). \end{aligned} \tag{31.29}$$

In magnetic problems one usually deals only with the set of $(2L + 1)(2S + 1)$ states determined by Hund's first two rules, all others lying so much higher in energy as to be of no interest. Furthermore, it is often enough to consider only the $2J + 1$ lowest lying of these specified by the third rule.

The rules are easier to apply than their description might suggest; indeed, in determining the lowest lying J-multiplet (known as a *term*) for ions in a solid, one really encounters only 22 cases of interest: 1 to 9 electrons in a d-shell ($l = 2$) or 1 to 13 electrons in an f-shell ($l = 3$).[19] For unfortunate historical reasons the ground-state multiplet in these cases is not described by the simple triad of numbers

[19] Partially filled p-shells contain valence electrons, and invariably broaden into bands in the solid. Thus the configuration of the electrons they contain in the solid is in no sense a slight distortion of the configuration in the free atom, and the analysis of this chapter is inapplicable.

SLJ. Instead, the orbital angular momentum L is given by a letter, according to the hallowed spectroscopic code:

$$\begin{aligned} L &= 0\ 1\ 2\ 3\ 4\ 5\ 6 \\ X &= S\ P\ D\ F\ G\ H\ I \end{aligned} \tag{31.30}$$

The spin is specified by affixing the number $2S + 1$ (known as the multiplicity) to the letter as a superprefix, and only J is given as the number J, affixed as a right subscript. Thus the lowest lying J-multiplet is described by the symbol: ${}^{(2S+1)}X_J$.

The cases of major interest for the study of magnetism in solids are given in Table 31.2.

Table 31.2
GROUND STATES OF IONS WITH PARTIALLY FILLED d- OR f-SHELLS, AS CONSTRUCTED FROM HUND'S RULES[a]

d-shell ($l = 2$)										
n	$l_z = 2,$	$1,$	$0,$	$-1,$	-2	S	$L = \lvert\Sigma l_z\rvert$	J		SYMBOL
1	↓					1/2	2	3/2	$J = \lvert L - S\rvert$	${}^2D_{3/2}$
2	↓	↓				1	3	2		${}^3F_2$
3	↓	↓	↓			3/2	3	3/2		${}^4F_{3/2}$
4	↓	↓	↓	↓		2	2	0		${}^5D_0$
5	↓	↓	↓	↓	↓	5/2	0	5/2		${}^6S_{5/2}$
6	⇅	↑	↑	↑	↑	2	2	4	$J = L + S$	${}^5D_4$
7	⇅	⇅	↑	↑	↑	3/2	3	9/2		${}^4F_{9/2}$
8	⇅	⇅	⇅	↑	↑	1	3	4		${}^3F_4$
9	⇅	⇅	⇅	⇅	↑	1/2	2	5/2		${}^2D_{5/2}$
10	⇅	⇅	⇅	⇅	⇅	0	0	0		${}^1S_0$

f-shell ($l = 3$)										
n	$l_z = 3,$	$2,$	$1,$	$0,$	$-1,$	$-2,$	-3	S	$L = \lvert\Sigma l_z\rvert$	J		
1	↓							1/2	3	5/2	$J = \lvert L - S\rvert$	${}^2F_{5/2}$
2	↓	↓						1	5	4		${}^3H_4$
3	↓	↓	↓					3/2	6	9/2		${}^4I_{9/2}$
4	↓	↓	↓	↓				2	6	4		${}^5I_4$
5	↓	↓	↓	↓	↓			5/2	5	5/2		${}^6H_{5/2}$
6	↓	↓	↓	↓	↓	↓		3	3	0		${}^7F_0$
7	↓	↓	↓	↓	↓	↓	↓	7/2	0	7/2		${}^8S_{7/2}$
8	⇅	↑	↑	↑	↑	↑	↑	3	3	6	$J = L + S$	${}^7F_6$
9	⇅	⇅	↑	↑	↑	↑	↑	5/2	5	15/2		${}^6H_{15/2}$
10	⇅	⇅	⇅	↑	↑	↑	↑	2	6	8		${}^5I_8$
11	⇅	⇅	⇅	⇅	↑	↑	↑	3/2	6	15/2		${}^4I_{15/2}$
12	⇅	⇅	⇅	⇅	⇅	↑	↑	1	5	6		${}^3H_6$
13	⇅	⇅	⇅	⇅	⇅	⇅	↑	1/2	3	7/2		${}^2F_{7/2}$
14	⇅	⇅	⇅	⇅	⇅	⇅	⇅	0	0	0		${}^1S_0$

[a] ↑ = spin $\frac{1}{2}$; ↓ = spin $-\frac{1}{2}$.

3d elements

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
$3d\,4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$	$3d^{10} 4s$	$3d^{10} 4s^2$

note the change:

No $3d^4 4s^2$ OR $3d^9 4s^2$

4f La — Lu

Generally, d-shell ($l = 2$)

# electrons	l_z (2 1 0 −1 −2)	S	$L = \lvert\Sigma l_z\rvert$	J	symbol
1	↑	$\frac{1}{2}$	2	$J=\lvert L-S\rvert$: $3/2$	$^2D_{3/2}$
2	↑ ↑	1	3	2	$^3F_2$
3	↑ ↑ ↑	$3/2$	3	$3/2$	$^4F_{3/2}$
4	↑ ↑ ↑ ↑	2	2	0	$^5D_0$
5	↑ ↑ ↑ ↑ ↑	$5/2$	0	$5/2$	$^6S_{5/2}$
6	↑↓ ↑ ↑ ↑ ↑	2	2	$J=L+S$: 4	$^5D_4$
7	↑↓ ↑↓ ↑ ↑ ↑	$3/2$	3	$9/2$	$^4F_{9/2}$
8	↑↓ ↑↓ ↑↓ ↑ ↑	1	3	4	$^3F_4$
9	↑↓ ↑↓ ↑↓ ↑↓ ↑	$1/2$	2	$5/2$	$^2D_{5/2}$
10	↑↓ ↑↓ ↑↓ ↑↓ ↑↓	0	0	0	$^1S_0$

Spectroscopic Symbol $^{2S+1}X_J$

X =	0	1	2	3	4	5	6
L =	S	P	D	F	G	H	I

Note that $n=4$ state has $J=0$ but $S=L=2 \Rightarrow$ Van Vleck paramagnet

Example : Copper

Cu $Z = 29$

Electronic configuration

$$1s^2\, 2s^2\, 2p^6\, 3s^2\, 3p^6\, 3d^{10}\, 4s$$

$s = (l = 0)$

$p = (l = 1)$

$d = (l = 2)$

$$Z = 2 + 2 + 6 + 2 + 6 + 10 + 1 = 29$$

In high-T_c superconductors Cu plays major role — exists in Cu^{2+} state $\Rightarrow$ $3d^9$ $\Rightarrow$ expect $\left.\begin{matrix} S = \frac{1}{2} \\ L = 2 \end{matrix}\right\}$ $J = L + S = \frac{5}{2}$

But in fact it is in $L = 0$, $S = \frac{1}{2}$ state due to <u>orbital quenching</u>